Hypothesis Testing
Differences Among Groups

II. Analysis of Variance (ANOVA)

C. Summary

III. Examples

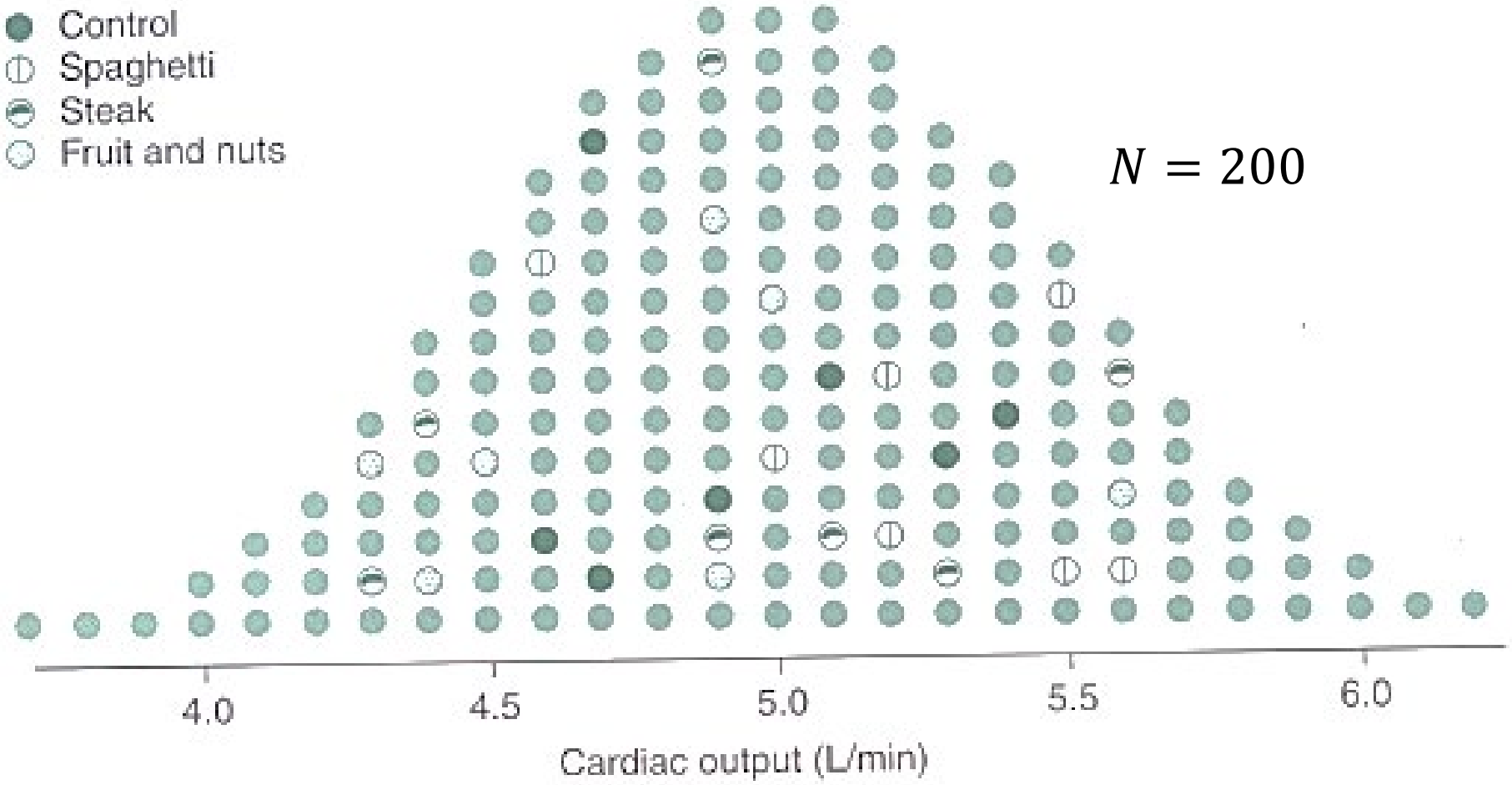
B. HR in BME 423
C. Lung Function and Tobacco-Pollutant Exposure

IV. Sum of Squares ANOVA Framework

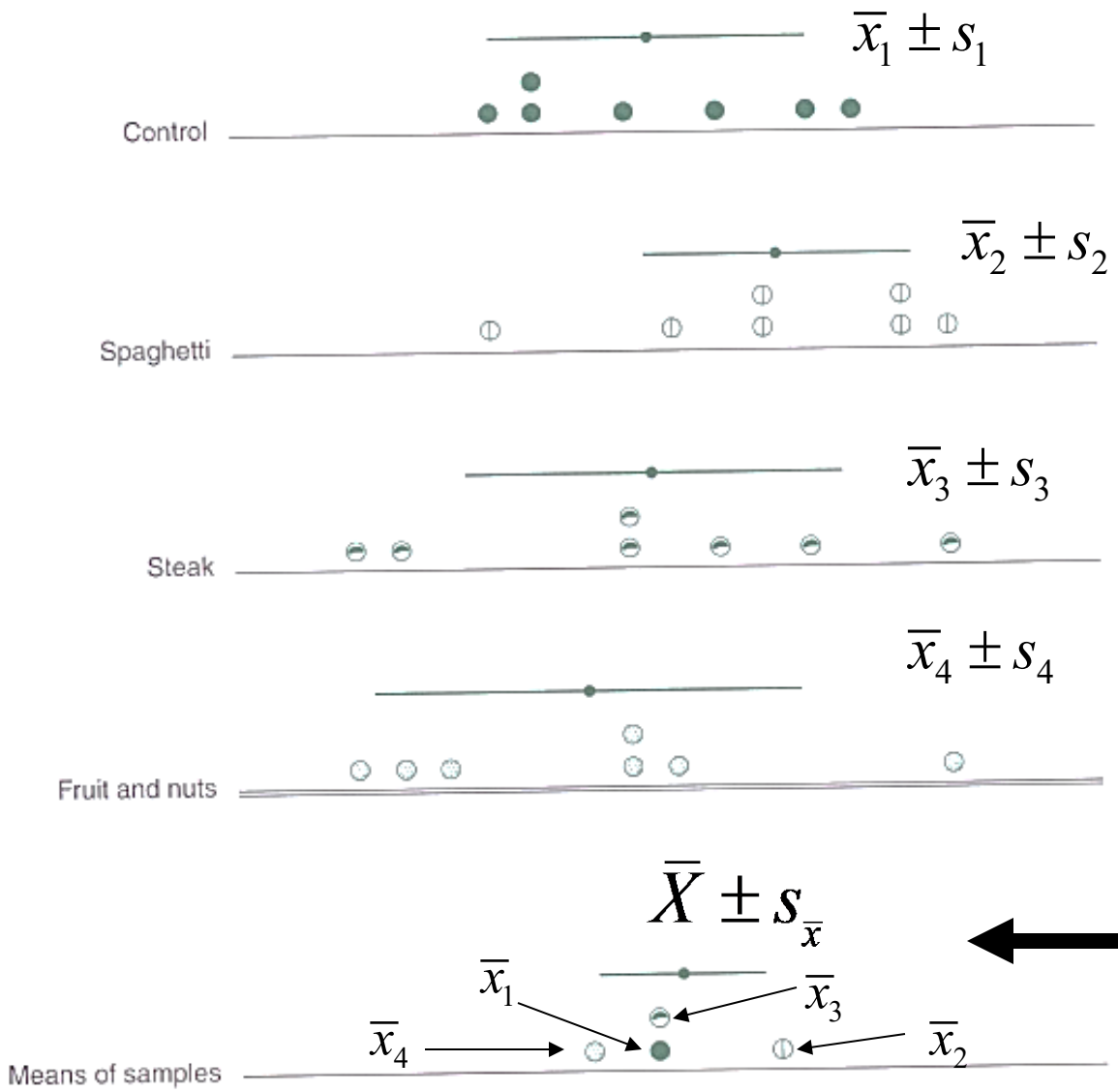
A. New Notation
B. Squared Deviation Formulation
C. Sum of Squares Examples

V. ANOVA in R

$$X \sim N(\mu, \sigma^2)$$



If we observe the entire population - Diet has No Effect on CO



Question:

Are these differences in CO due to diet or are these due to random variation in CO in the population?

Is this variability in $\bar{x}_j$ consistent with the variability observed within the individual sample x_i values?

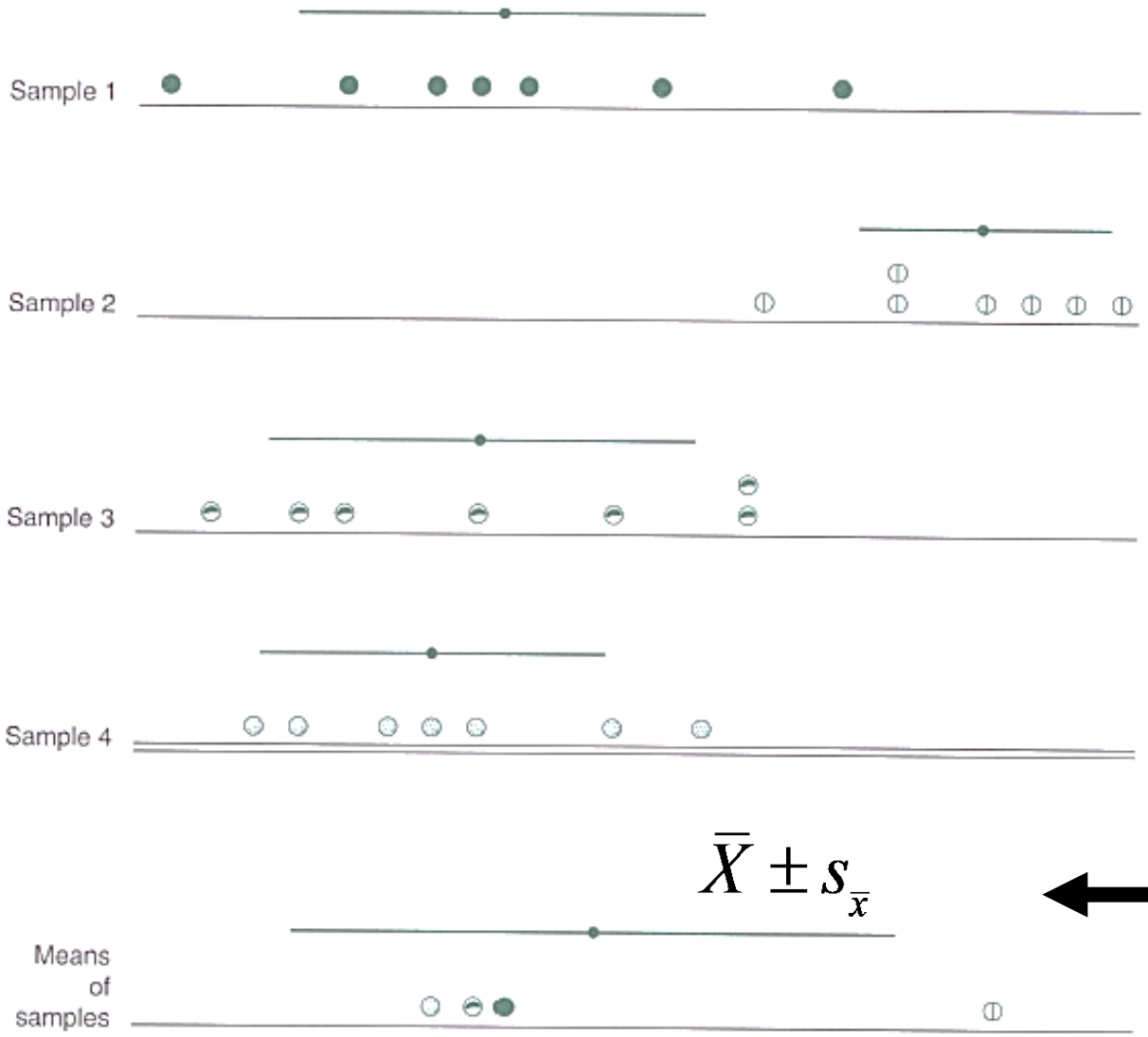


FIGURE 3-3. The four samples shown are identical to those in Figure 3-2 except that the variability in the mean values has been increased substantially. The samples now appear to differ from each other because the variability between the sample means is larger than one would expect from the variability within each sample. Compare the relative variability in mean values with the variability within the sample groups with that seen in Figure 3-2.

Is this variability in $\bar{x}_j$ consistent with the variability observed within the individual sample x_i values?

II. ANOVA: C. Summary

Experiment:

RV X is measured

m - # of treatment groups
n - # number of subjects per group

Null Hypothesis:

H_0: $\mu_1 = \mu_2 = \ldots = \mu_m$ Alternative hypoth: at least 2 $\mu's$ differ

1. Mean of each group

$$\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_m \qquad \left(\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i \right)$$

2. Variance of each group

$$s_1^2, s_2^2, \ldots, s_m^2 \qquad \left(s^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(x_i - \bar{x} \right)^2 \right)$$

3. Calculate the within-groups est. of variance

$$s_{wit}^2 = \frac{1}{m}\sum_{i=1}^{m} s_i^2$$

4. Compute the overall (grand) mean

$$\bar{X} = \frac{1}{m}\sum_{i=1}^{m} \bar{x}_i$$

5. Compute standard error of the mean

$$s_{\bar{x}}^2 = \frac{1}{m-1}\sum_{i=1}^{m}\left(\bar{x}_i - \bar{X}\right)^2$$

6. Calculate the between-groups est. of variance

$$s_{bet}^2 = n s_{\bar{x}}^2$$

7. Compute the F Statistic

$$F = \frac{s_{bet}^2}{s_{wit}^2}$$

$\nu_n = m - 1$ Numerator degrees of freedom

$\nu_d = m(n-1)$ Denominator degrees of freedom

8. Determine F_{crit} (critical value of F)

$$F_{crit} \equiv F_{0.05}(\nu_n, \nu_d)$$ Obtained from F Table

In general $F_{crit} \equiv F_{\alpha}(\nu_n, \nu_d)$ α: significance level

9. Compare F to F_{crit} and State Conclusion

If $F > F_{crit}$ Reject null hypothesis with $P < \alpha$

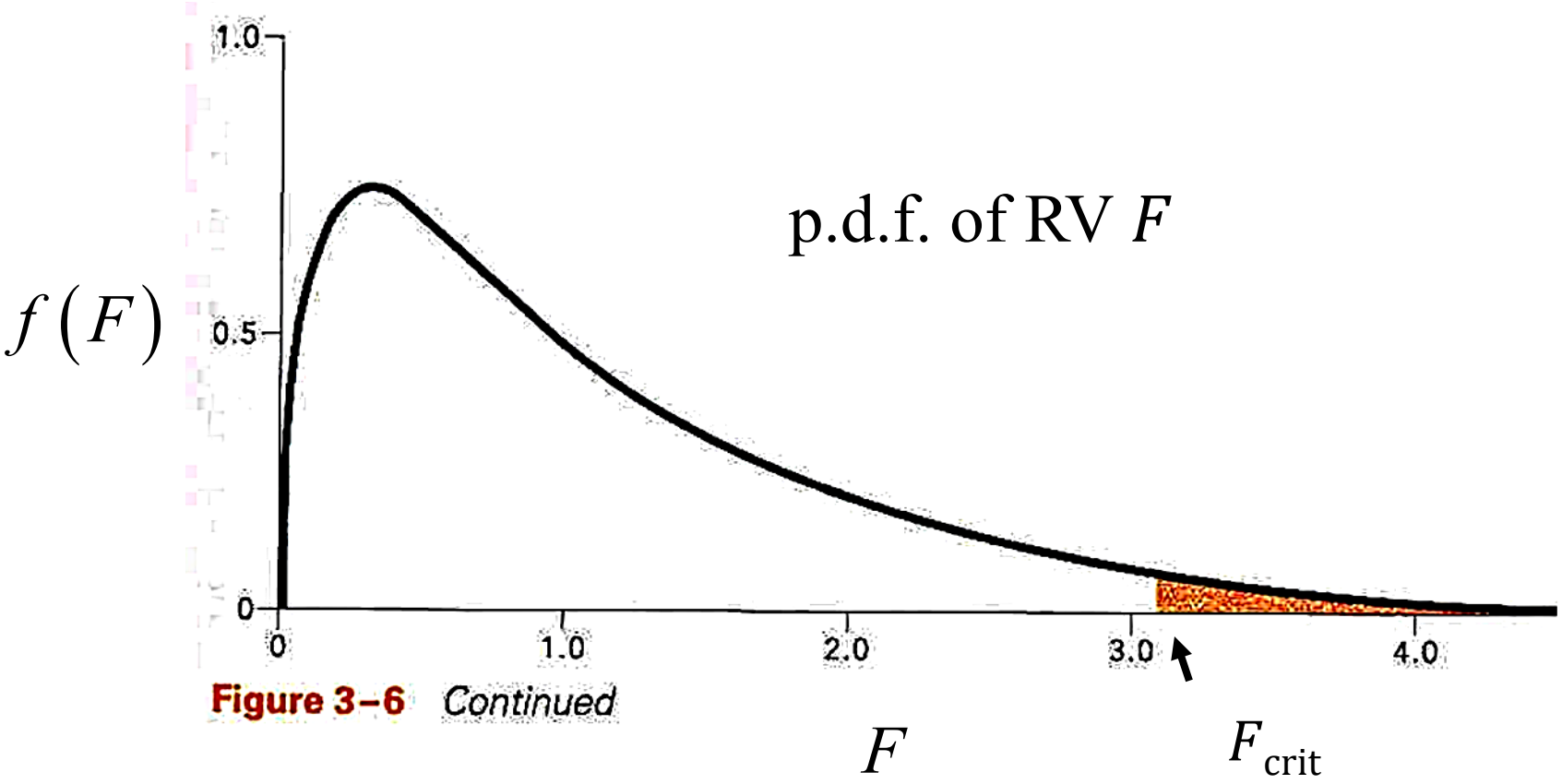


Figure 3–6 *Continued*

Assumptions of ANOVA

1. Data are interval data
2. Samples must be independent
3. Samples must be randomly selected
4. Population(s) must be Normally distributed
5. Variance(s) of each population must be equal

III Examples: B. HR in BME 423

Experiment (Observations):

Measure HR in a sample of BME 423 students

$m = 2$ (pet choice: Cat or Dog)
$n = 5$ subjects per group

Null Hypothesis:

H_0: No difference in HR between students who prefer cats versus those who prefer dogs

1. Mean of samples from each group

$\bar{x}_C = 74.2$ $\bar{x}_D = 76.8$

2. Variance of sample from each group

${s_C}^2 = (7.3)^2$ ${s_D}^2 = (9.1)^2$

units:
BPM for means
$(BPM)^2$ for variances

3. Calculate the within-groups est. of variance

$$s_{wit}^2 = \frac{1}{m}\sum_{i=1}^{m} s_i^2$$

$$s_{wit}{}^2 = \frac{1}{2}(s_C{}^2 + s_D{}^2) = 68.05$$

4. Compute the overall (grand) mean

$$\bar{X} = \frac{1}{m}\sum_{i=1}^{m} \bar{x}_i$$

5. Compute standard error of the mean

$$s_{\bar{x}}^2 = \frac{1}{m-1}\sum_{i=1}^{m}\left(\bar{x}_i - \bar{X}\right)^2$$

6. Calculate the between-groups est. of variance

$$s_{bet}^2 = n s_{\bar{x}}^2$$

3. Calculate the within-groups est. of variance

$$s_{wit}^2 = \frac{1}{m}\sum_{i=1}^{m} s_i^2$$

4. Compute the overall (grand) mean

$$\bar{X} = \frac{1}{m}\sum_{i=1}^{m} \bar{x}_i$$

$$\bar{X} = \frac{1}{2}(\bar{x}_C + \bar{x}_D) = 75.5$$

5. Compute standard error of the mean

$$s_{\bar{x}}^2 = \frac{1}{m-1}\sum_{i=1}^{m}\left(\bar{x}_i - \bar{X}\right)^2$$

6. Calculate the between-groups est. of variance

$$s_{bet}^2 = n s_{\bar{x}}^2$$

3. Calculate the within-groups est. of variance

$$s_{wit}^2 = \frac{1}{m}\sum_{i=1}^{m} s_i^2$$

4. Compute the overall (grand) mean

$$\bar{X} = \frac{1}{m}\sum_{i=1}^{m} \bar{x}_i$$

5. Compute standard error of the mean

$$s_{\bar{x}}^2 = \frac{1}{m-1}\sum_{i=1}^{m}\left(\bar{x}_i - \bar{X}\right)^2$$

$$s_{\bar{x}}{}^2 = \frac{(\bar{x}_C - 75.5)^2 + (\bar{x}_D - 75.5)^2}{2-1} = 3.38$$

6. Calculate the between-groups est. of variance

$$s_{bet}^2 = n s_{\bar{x}}^2$$

3. Calculate the within-groups est. of variance

$$s_{wit}^2 = \frac{1}{m}\sum_{i=1}^{m} s_i^2$$

4. Compute the overall (grand) mean

$$\bar{X} = \frac{1}{m}\sum_{i=1}^{m} \bar{x}_i$$

5. Compute standard error of the mean

$$s_{\bar{x}}^2 = \frac{1}{m-1}\sum_{i=1}^{m}\left(\bar{x}_i - \bar{X}\right)^2$$

6. Calculate the between-groups est. of variance

$$s_{bet}^2 = n s_{\bar{x}}^2$$

$$s_{bet}{}^2 = 5 \cdot 3.38 = 16.9$$

7. Compute the F Statistic

$$F = \frac{s_{bet}^2}{s_{wit}^2}$$

$$\nu_n = m - 1$$

$$\nu_d = m(n-1)$$

8. Compare F with the F_{crit} (critical value of F)

$$F_{crit} \equiv F_\alpha(\nu_n, \nu_d)$$

Obtained from F Table

α: significance level; acceptable probability of a false positive

9. State Conclusion

$$F > F_{crit}$$

7. Compute the F Statistic

$$F = \frac{{s_{bet}}^2}{{s_{wit}}^2} = \frac{16.9}{68.05} = 0.248$$

$$\nu_n = 2 - 1 = 1$$

$$\nu_d = 2(5 - 1) = 8$$

8. Compare F with the F_{crit} (critical value of F)

$$F_{crit} \equiv F_{\alpha}(\nu_n, \nu_d)$$

Obtained from F Table

α: significance level; acceptable probability of a false positive

9. State Conclusion

$$F > F_{crit}$$

7. Compute the F Statistic

$$F = \frac{{s_{bet}}^2}{{s_{wit}}^2} = \frac{16.9}{68.05} = 0.248$$

$$\nu_n = 2 - 1 = 1$$

$$\nu_d = 2(5 - 1) = 8$$

8. Compare F with the F_{crit} (critical value of F)

$$F_{crit} \equiv F_{0.05}(1,8) = 5.32$$

Obtained from F Table

α: significance level; acceptable probability of a false positive

9. State Conclusion

$$F > F_{crit}$$

7. Compute the F Statistic

$$F = \frac{{s_{bet}}^2}{{s_{wit}}^2} = \frac{16.9}{68.05} = 0.248$$

$$\nu_n = 2 - 1 = 1$$

$$\nu_d = 2(5 - 1) = 8$$

8. Compare F with the F_{crit} (critical value of F)

$$F_{crit} \equiv F_{0.05}(1,8) = 5.32$$ Obtained from F Table

α: significance level; acceptable probability of a false positive

9. State Conclusion

$$F = 0.248 \not\geq F_{crit} = 5.32$$

Cannot reject null hypothesis; conclude no difference in HR between the BME 423 students who prefer cats and those who prefer dogs

III Examples: C. Lung Function and Tobacco Pollutant Exposure

TABLE 3-5. Mean Forced Midexpiratory Flow (L/s)

Group	Sample Size (n)	Mean	Standard Deviation
Nonsmokers			
Worked in smoke-free environment	200	3.17	0.74
Worked in smoky environment	200	2.72	0.71
Light smokers	200	2.63	0.73
Moderate smokers	200	2.29	0.70
Heavy smokers	200	2.12	0.72

- Maximal expiratory flow-volume (MEFV) is a way of measuring lung functionality (measures how rapidly a person can force air from his/her lungs)
- Reduced mean forced mid-expiratory flow (the mid-portion of MEFV) is associated with small-airways disease of the lungs
- 5 groups ($m = 5$) of varying tobacco pollutant exposure with 200 subjects in each group ($n = 200$); does lung functionality differ among these groups?
- Null Hypothesis H_0: No difference in measured lung function among these 5 tobacco-exposure groups

3. Calculate the within-groups est. of variance

$$s_{wit}^2 = \frac{1}{m}\sum_{i=1}^{m} s_i^2$$

$$s_{wit}^2 = \frac{1}{5}\left(s_{SF}^2 + s_{SE}^2 + s_{LS}^2 + s_{MS}^2 + s_{HS}^2\right) = 0.52$$

4. Compute the overall (grand) mean

$$\bar{X} = \frac{1}{m}\sum_{i=1}^{m} \bar{x}_i$$

5. Compute standard error of the mean

$$s_{\bar{x}}^2 = \frac{1}{m-1}\sum_{i=1}^{m}\left(\bar{x}_i - \bar{X}\right)^2$$

6. Calculate the between-groups est. of variance

$$s_{bet}^2 = n s_{\bar{x}}^2$$

units:
L/s for means
$(L/s)^2$ for variances

3. Calculate the within-groups est. of variance

$$s^2_{wit} = \frac{1}{m}\sum_{i=1}^{m} s_i^2$$

4. Compute the overall (grand) mean

$$\bar{X} = \frac{1}{m}\sum_{i=1}^{m} \bar{x}_i$$

$$\bar{X} = \frac{1}{5}\left(\bar{x}_{SF} + \bar{x}_{SE} + \bar{x}_{LS} + \bar{x}_{MS} + \bar{x}_{HS}\right) = 2.59$$

5. Compute standard error of the mean

$$s^2_{\bar{x}} = \frac{1}{m-1}\sum_{i=1}^{m}\left(\bar{x}_i - \bar{X}\right)^2$$

6. Calculate the between-groups est. of variance

$$s^2_{bet} = n s^2_{\bar{x}}$$

3. Calculate the within-groups est. of variance

$$s^2_{wit} = \frac{1}{m}\sum_{i=1}^{m} s_i^2$$

4. Compute the overall (grand) mean

$$\bar{X} = \frac{1}{m}\sum_{i=1}^{m} \bar{x}_i$$

5. Compute standard error of the mean

$$s^2_{\bar{x}} = \frac{1}{m-1}\sum_{i=1}^{m}\left(\bar{x}_i - \bar{X}\right)^2$$

$$s^2_{\bar{x}} = \frac{\left(\bar{x}_{SF} - \bar{X}\right)^2 + \left(\bar{x}_{SE} - \bar{X}\right)^2 + \left(\bar{x}_{LS} - \bar{X}\right)^2 + \left(\bar{x}_{MS} - \bar{X}\right)^2 + \left(\bar{x}_{HS} - \bar{X}\right)^2}{m-1} = 0.166$$

6. Calculate the between-groups est. of variance

$$s^2_{bet} = ns^2_{\bar{x}}$$

3. Calculate the within-groups est. of variance

$$s_{wit}^2 = \frac{1}{m}\sum_{i=1}^{m} s_i^2$$

4. Compute the overall (grand) mean

$$\bar{X} = \frac{1}{m}\sum_{i=1}^{m} \bar{x}_i$$

5. Compute standard error of the mean

$$s_{\bar{x}}^2 = \frac{1}{m-1}\sum_{i=1}^{m}\left(\bar{x}_i - \bar{X}\right)^2$$

6. Calculate the between-groups est. of variance

$$s_{bet}^2 = ns_{\bar{x}}^2$$

$$s_{bet}^2 = ns_{\bar{x}}^2 = 33.3$$

7. Compute the F Statistic

$$F = \frac{s_{bet}^2}{s_{wit}^2} = \frac{33.3}{0.52} = 64.18$$

$$\nu_n = m - 1 = 4$$

$$\nu_d = m(n-1) = 995$$

8. Compare F with the F_{crit} (critical value of F)

$$F_{crit}$$

Obtained from F Table

α: significance level; acceptable probability of a false positive

9. State Conclusion

$$F > F_{crit}$$

7. Compute the F Statistic

$$F = \frac{s^2_{bet}}{s^2_{wit}} = \frac{33.3}{0.52} = 64.18$$

$$\nu_n = m - 1 = 4$$

$$\nu_d = m(n-1) = 995$$

8. Compare F with the F_{crit} (critical value of F)

$$F_{0.01}(4,\infty) = 3.32 < F_{crit} \equiv F_{0.01}(4,995) < 3.48 = F_{0.01}(4,120)$$

Obtained from F Table

9. State Conclusion

$$F > F_{crit}$$

7. Compute the F Statistic

$$F = \frac{s_{bet}^2}{s_{wit}^2} = \frac{33.3}{0.52} = 64.18$$

$$\nu_n = m - 1 = 4$$

$$\nu_d = m(n-1) = 995$$

8. Compare F with the F_{crit} (critical value of F)

$$F_{0.01}(4,\infty) = 3.32 < F_{crit} \equiv F_{0.01}(4,995) < 3.48 = F_{0.01}(4,120)$$

Obtained from F Table

9. State Conclusion

$$F = 64.18 > 3.48 > F_{crit}$$

Reject null hypothesis and conclude lung function differs among tobacco-exposure groups with $P < 0.01$

IV. Sum of Squares ANOVA Framework

A. New Notation

Note: In Chapter 9 only the equal # of subjects per group case is presented

Table 9-2 Notation for One-Way Analysis of Variance

Treatment			
1	2	3	4
X_{11}	X_{21}	X_{31}	X_{41}
X_{12}	X_{22}	X_{32}	X_{42}
X_{13}	X_{23}	X_{33}	X_{43}
X_{14}		X_{34}	X_{44}
			X_{45}
			X_{46}
			X_{47}

Number of subjects in each group does not need to be the same for this formulation

x_{ts} t – treatment group $t = 1, \ldots, m$

s – subject number $s = 1, \ldots, n_t$

B. Squared Deviation Formulation

1. Mean of each group

$$\bar{x}_1,\ \bar{x}_2,\ \ldots,\bar{x}_m \qquad \left(\bar{x}_t = \frac{1}{n_t}\sum_{s=1}^{n_t} x_{ts} \right)$$

2. Sum of squared deviations for each group

$$SS_1,\ SS_2,\ \ldots,SS_m \qquad \left(SS_t = \sum_{s=1}^{n_t}\left(x_{ts} - \bar{x}_t\right)^2 \right)$$

Note: previously, with an equal # in each group, we calculated group variances

$$s_1^2,\ s_2^2,\ \ldots,s_m^2 \qquad \left(s^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2 \right)$$

3. Calculate the **within-groups** sum of squares

$$SS_{wit} = \sum_{t=1}^{m} SS_t$$

4. Calculate the **within-groups** est. of variance

Mean Squared (MS) deviation

$$s^2_{wit} = \frac{SS_{wit}}{DF_{wit}} \equiv MS_{wit}$$

$$DF_{wit} = N - m$$

$$N = \sum_{t=1}^{m} n_t = \text{total number of subjects in whole experiment}$$

Degrees of freedom

5. Calculate the **between-groups** sum of squares

Deviation of $\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_m$ from $\overline{X}$ (the grand mean)

$$SS_{bet} = \sum_{t=1}^{m} n_t \bar{x}_t^2 - \frac{\left(\sum_{t=1}^{m} n_t \bar{x}_t \right)^2}{N}$$

6. Calculate the **between-groups** est. of variance

$$s_{bet}^2 = \frac{SS_{bet}}{DF_{bet}} \equiv MS_{bet} \qquad DF_{bet} = m - 1$$

7. Compute the F Statistic

$$F = \frac{s_{bet}^2}{s_{wit}^2}$$

$\nu_n = DF_{bet} = m - 1$ Numerator degrees of freedom

$\nu_d = DF_{wit} = N - m$ Denominator degrees of freedom

8. Compare F with the F_{crit} (critical value of F)

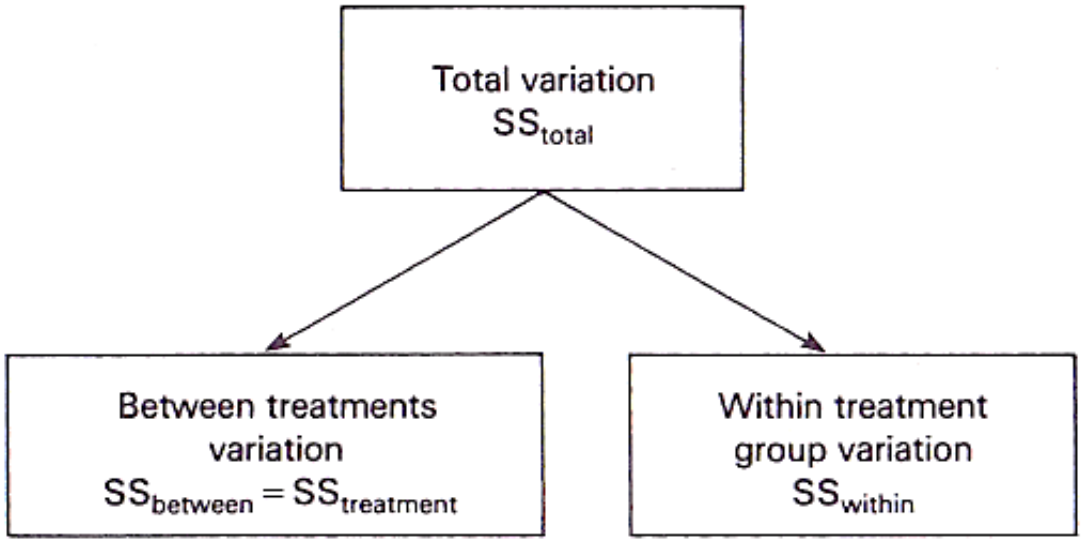
$$F_{crit} \equiv F_\alpha(\nu_n, \nu_d)$$

α: significance level; acceptable probability of a false positive

9. State Conclusion

If $F > F_{crit}$ Reject null hypothesis with $P < \alpha$

ANOVA – Squared Error Formulation



$$\left(SS_t = \sum_{s=1}^{n_t} \left(x_{ts} - \bar{x}_t \right)^2 \right)$$

$$SS_{bet} = \sum_{t=1}^{m} n_t \bar{x}_t^2 - \frac{\left(\sum_{t=1}^{m} n_t \bar{x}_t \right)^2}{N}$$

$$SS_{wit} = \sum_{t=1}^{m} SS_t$$

$$s_{bet}^2 = \frac{SS_{bet}}{DF_{bet}} \equiv MS_{bet}$$

$$s_{wit}^2 = \frac{SS_{wit}}{DF_{wit}} \equiv MS_{wit}$$

$$F = \frac{s_{bet}^2}{s_{wit}^2} = \frac{MS_{bet}}{MS_{wit}} \qquad \nu_n = DF_{bet} = m - 1 \qquad \nu_d = DF_{wit} = N - m$$

C. Sum of Squares Example: Cardiac Output

Table 9-1 Cardiac Output (L/min) in Different Groups of Seven People Fed Different Diets

	Treatment			
	Control	Spaghetti	Steak	Fruit and nuts
	4.6	4.6	4.3	4.3
	4.7	5.0	4.4	4.4
	4.7	5.2	4.9	4.5
	4.9	5.2	4.9	4.9
	5.1	5.5	5.1	4.9
	5.3	5.5	5.3	5.0
	5.4	5.6	5.6	5.6
Treatment (column) means	4.96	5.23	4.93	4.80
Treatment (column) sums of squares	0.597	0.734	1.294	1.200
Grand mean = 4.98		Total sum of squares = 4.501		

$$\left(SS_t = \sum_{s=1}^{n_t} \left(x_{ts} - \bar{x}_t \right)^2 \right)$$

3. Calculate the **within-groups** sum of squares

$$SS_{wit} = \sum_{t=1}^{4} SS_t = 3.825$$

4. Calculate the **within-groups** est. of variance

$$s^2_{wit} = \frac{SS_{wit}}{DF_{wit}} = \frac{3.825}{24} = 0.159$$

$$DF_{wit} = N - m = 28 - 4 = 24$$

$$N = \sum_{t=1}^{4} n_t = \sum_{t=1}^{4} 7 = 28$$

5. Calculate the **between-groups** sum of squares

$$SS_{bet} = \sum_{t=1}^{m} n_t \bar{x}_t^2 - \frac{\left(\sum_{t=1}^{m} n_t \bar{x}_t \right)^2}{N}$$

$$SS_{bet} = \sum_{t=1}^{4} 7\bar{x}_t^2 - \frac{\left(\sum_{t=1}^{4} 7\bar{x}_t \right)^2}{28} = 0.685$$

6. Calculate the **between-groups** est. of variance

$$s_{bet}^2 = \frac{SS_{bet}}{DF_{bet}} = \frac{0.685}{3} = 0.228 \qquad DF_{bet} = m - 1 = 3$$

7. Compute the *F* Statistic

$$F = \frac{s^2_{bet}}{s^2_{wit}} = \frac{0.228}{0.159} = 1.43$$

$$\nu_n = DF_{bet} = 3$$
$$\nu_d = DF_{wit} = 24$$

8. Conclusion

$$F = 1.43 < F_{0.05}(3, 24) = 3.01$$

CANNOT reject null hypothesis.
Conclude no effect of these diets on CO.

Guidelines for which ANOVA Formulation to Use

- If number of subjects in each group is different, must use sum of squares formulation (requires raw data)

- If number of subjects in each group is same (balanced study design), and have raw data, can use either formulation

- If design is balanced and only have means and standard deviations for each group, must use "comparison of variance" formulation

C. Sum of Squares Example: Nitric Oxide

Measurements of exhaled nitric oxide (ppb) from 22 young asthmatic subjects following exposure to an allergen are shown below. These subjects were divided into 3 groups according to lung function tests. The researchers in this study hypothesized that there would be no difference in exhaled nitric oxide in these three lung function groups. Were they correct? You can assume the data are Normally distributed.

Group A	Group B	Group C
23.1	9.1	10.5
37.8	10.8	16.1
19.4	6.5	24.5
29.3	21.0	12.4
41.0	18.4	20.6
26.5	23.9	18.6
	17.7	
	34.8	
	4.5	
	10.1	

One-Way ANOVA in R

```
aov(formula = response ~ factor, data = data_name)
```

- **formula**: tells R how it will be analyzing the data (i.e., specifies relationship between/among variables involved)
- ***response***: outcome variable being modeled (i.e., data being analyzed)
- ***factor***: variable that separates data into groups (can be thought of as "predictor" variable)
- **data**: tells R where variables are stored
- ***data_name***: name of data frame that contains response and factor variables

Nitric Oxide Example Done in R

```
> my.aov <- aov(formula = Exhaled_NO ~ Group, data = NO_data)
> print(my.aov)
Call:
   aov(formula = Exhaled_NO ~ Group, data = NO_data)

Terms:
                    Group Residuals
Sum of Squares   779.3819 1269.4927
Deg. of Freedom         2        19

Residual standard error: 8.174069
Estimated effects may be unbalanced
> summary(my.aov)
            Df Sum Sq Mean Sq F value Pr(>F)
Group        2  779.4   389.7   5.832 0.0106 *
Residuals   19 1269.5    66.8
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

"Group": between-groups measures
"Residuals": within-groups measures